Date: **Class:** **Topic:**

xx-yy-zzzz | Science 101 | Some more examples for supported markdown structures

You can create/modify your own header styles too.

Top and bottom heights are limited (lower limits is 5mm):

- Top-Limit up to: 80mm ~ 3.15″
  - usually used as header
- Bottom-Limit up to: 100mm ~ 3.94″
  - usually used as summary

Chemistry Formulas:

$Fe^{II}Fe^{III}{}_2O_4$

$A \xrightarrow{H_2O} B$

$A{-}B{=}C{\equiv}D$ chem. bonding

$C_p[H_2O\,(l)] = 75.3\ \frac{J}{mol\,K}$

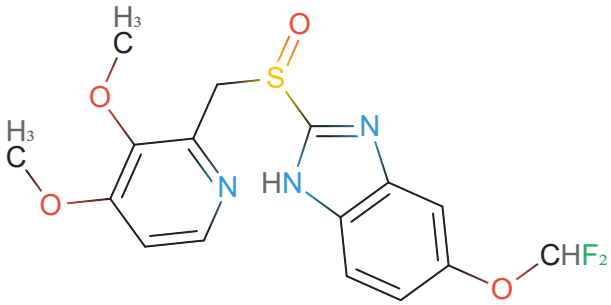
Math formula sizing by using standard $\LaTeX$ syntax:

1. `\tiny` – ca. **5–6 pt**
2. `\scriptsize` – ca. **7 pt**
3. `\small` – ca. **9 pt**
4. `\normalsize` – **10–12 pt** (not needed!)
5. `\large` – ca. **12–14 pt**
6. `\Large` – ca. **14–16 pt**
7. `\LARGE` – ca. **17–20 pt**
8. `\huge` – ca. **20–24 pt**
9. `\Huge` – ca. **25–30 pt** in MathJax.

Chemical Reaction Equations with mhchem Package: mhchem Syntax

$$Zn^{2+} \underset{+2\,H^+}{\overset{+2\,OH^-}{\rightleftharpoons}} \underset{\text{amphoteres Hydroxid}}{Zn(OH)_2\downarrow} \underset{+2\,H^+}{\overset{+2\,OH^-}{\rightleftharpoons}} \underset{\text{Hydroxozikat}}{[Zn(OH)_4]^{2-}}$$

Molecules with SMILES:



The molecular formula of Pantoprazol is created with $\LaTeX$ code and the mhchem package: $C_{16}H_{15}F_2N_3O_4S$.

The molecule pic was generated using SMILES notation code. Simply search for "Pantoprazol molecule" online to find the code for both - no need to create it yourself. Copy & paste - it's done! For not so well known molecules ask your LLM of choice.

For beginners in bioinformatics, chemistry, biology, and medicine:
You might wonder why hydrogen atoms (H) are often missing in SMILES notation? Here's the why: SMILES simplifies molecular structures by omitting atoms. While hydrogen is commonly implicit, other atoms like carbon, nitrogen, oxygen, sulfur, halogens, and even metals can also be omitted or simplified. The specific atoms that can be implicit depend on conventions used in the chemical database or software interpreting the notation. Understanding these conventions is crucial for accurately interpreting SMILES strings. Generating SMILES strings for specific molecules isn't always easy, but you can get them online from databases, search engines or you can ask a Large Language Model (LLM) to help. So you really don't have to do it by yourself.

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5. large-size:

$$\frac{\partial u}{\partial t} = p^2 \left(\frac{\partial^2 u}{\partial x^2} + \frac{\partial^2 u}{\partial y^2} + \frac{\partial^2 u}{\partial z^2} \right)$$

6. Large-size:

$$\frac{\partial u}{\partial t} = p^2 \left(\frac{\partial^2 u}{\partial x^2} + \frac{\partial^2 u}{\partial y^2} + \frac{\partial^2 u}{\partial z^2} \right)$$

7. LARGE-size:

$$\frac{\partial u}{\partial t} = p^2 \left(\frac{\partial^2 u}{\partial x^2} + \frac{\partial^2 u}{\partial y^2} + \frac{\partial^2 u}{\partial z^2} \right)$$

8. huge-size:

$$\frac{\partial u}{\partial t} = p^2 \left(\frac{\partial^2 u}{\partial x^2} + \frac{\partial^2 u}{\partial y^2} + \frac{\partial^2 u}{\partial z^2} \right)$$

9. Huge-size:

$$\frac{\partial u}{\partial t} = p^2 \left(\frac{\partial^2 u}{\partial x^2} + \frac{\partial^2 u}{\partial y^2} + \frac{\partial^2 u}{\partial z^2} \right)$$

1. Tiny:

$\begin{vmatrix} a & b \\ c & d \end{vmatrix} = (ad - bc)$

2. Script-size:

$\begin{vmatrix} a & b \\ c & d \end{vmatrix} = (ad - bc)$

3. Small-size:

$\sum_{i=1}^{n} i^2 = \frac{1}{2}n(n+1)$

4. Standard-size:

$\sum_{i=1}^{n} i^2 = \frac{1}{2}n(n+1)$

Lorem ipsum dolor sit amet, consetetur sadipscing elitr, sed diam nonumy eirmod tempor invidunt ut labore et dolore magna aliquyam erat, sed diam voluptua. Stet clita kasd gubergren, no sea takimata sanctus est Lorem ipsum dolor sit amet. Lorem ipsum dolor sit amet, consetetur sadipscing elitr, sed diam nonumy eirmod tempor invidunt ut labore et dolore magna aliquyam erat, sed diam voluptua.

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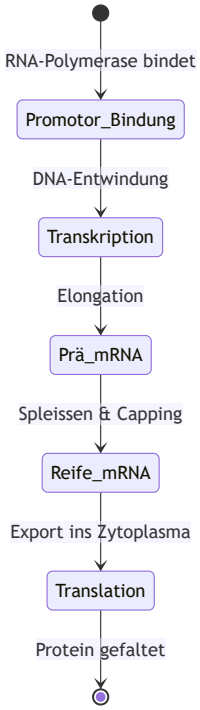
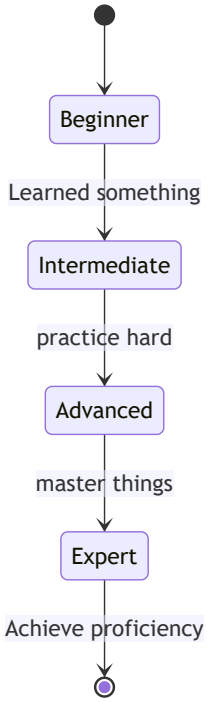
Tag-Pills for highlighting keywords:

#chemistry #biology #math #topics





My custom callout heading

- Lorem ipsum dolor sit
  - consetetur sadipscing elitr

- 1 custom type (choose Title and color with hex) syntax: **> [!custom|#1a20d9] My custom callout...**
- All 12 basic callout types (not the original Obsidian callout boxes, so with limited functionality)

Note
invidunt ut labore et dolore aliquyam erat

Abstract
invidunt ut labore et dolore aliquyam erat

Info
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Failure
invidunt ut labore et dolore aliquyam erat

Tip
invidunt ut labore et dolore aliquyam erat

Danger
invidunt ut labore et dolore aliquyam erat

Success
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Bug
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Question
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Example
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Warning
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Quote
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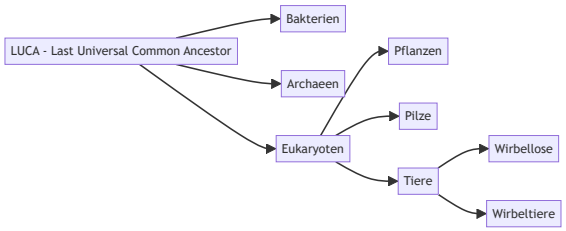
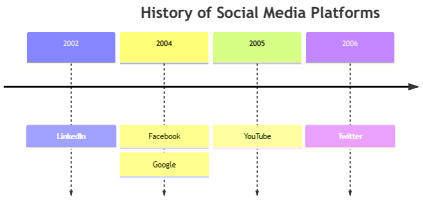
Some Mermaid Diagramms:

The syntax is quite easy to learn, but there are also graphical tools available and of course LLM's. Several software tools offer a **graphical user interface (GUI)** for creating Mermaid diagrams and exporting them as Mermaid code: **Mermaid Flow**, **Mermaid Chart**, **Miro** and **Eraser**